6 **(a)** Electrons in a vacuum are accelerated from rest through a potential difference (p.d.) V to form a beam. Each electron has mass m and charge q.

The beam is incident on a graphite crystal that acts as a diffraction grating. After passing through the crystal, the beam reaches a fluorescent screen. An interference pattern is observed on this screen.

(i) Explain what this observation shows about the nature of electrons.

.....

..... [1]

(ii) Determine an expression, in terms of m, q, V, for the momentum p of an electron in the beam.

p = ... [2]

(iii) The p.d. through which the electrons are accelerated is now increased to a greater value. Describe and explain the effect of this change on the interference pattern observed.

.....

.....

.....

.....

..... [2]

(b) A spherical quantum dot made of a semiconductor can be modelled as an electron confined in a one-dimensional infinite square well of width L as shown in Fig. 6.1. L is taken to be 5.0 nm.

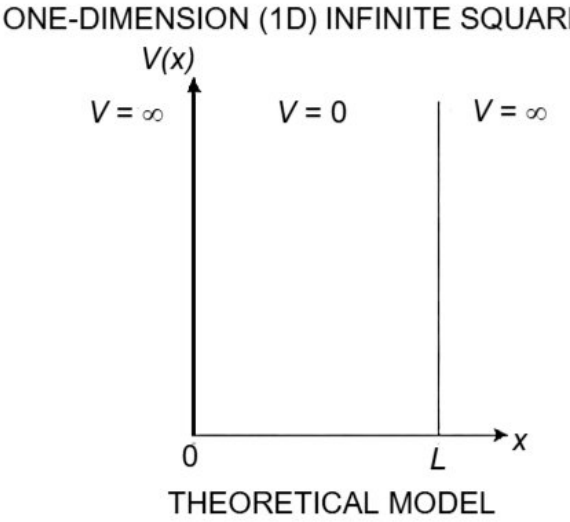


Fig. 6.1

The normalised ground-state wavefunction of the electron inside the well ($0 < x < L$) is:

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right)$$

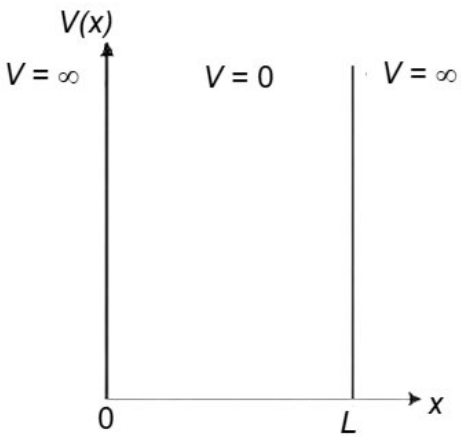
Outside the well, $\psi_1(x) = 0$.

(i) Explain what is meant a normalised wavefunction.

.....

..... [1]

(ii) Sketch the wavefunction in Fig. 6.2.



[1]

Fig. 6.2

(iii) Calculate the ground state energy E_1 of the electron in this quantum dot.

energy = ... J [1]

(iv) If the electron makes a transition from $n = 2$ to $n = 1$, find the wavelength of the emitted photon.

wavelength = ... m [2]

[Total: 10]